

TIME LEFT: 00:24:25

21 ★ QUESTION: 22 GROUP: Programming Basics SECTION: Error Finding Mark(s): 1

22 ☆ Identify the error in the given code snippet:

```
#include <stdio.h>

int main() {
    int x = 5;
    int y = 10;
    swap(&x, &y);
    printf("After swapping: x = %d, y = %d\n", x, y);
    return 0;
}

void swap(int *a, int *b) {
    int temp = *a;
    *a = *b;
    *b = temp;
}
```

☐ Missing return statement in the main function☐ Incorrect use of pointers☐ Incorrect use of pointers

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21 ★ QUESTION: 23 GROUP: Programming Basics SECTION: C Programming Mark(s): 1

22 Predict the output or error(s) if any for the code given below.

```
#include <stdio.h>
#define a 10
int main()
{
    #define a 50
    printf("%d", a);
    return 0;
}
```

☐ Nothing will be printed☐ 50☐ 10☐ Compile time error

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QUESTION: 24 GROUP: Programming Basics SECTION: Error Finding Mark(s): 1

Identify the line number that contains an error (if any):

```
#include <stdio.h>
int main()           //Line 1
{
    printf("Hello, world!\n"); // Line 3
    return 0;         //Line 4
}
```

☐ Line 4

☐ No Error

☐ Line 1

☐ Line 3

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QUESTION: 25 GROUP: Programming Basics SECTION: CA Mark(s): 1

Convert the following decimal number into hexadecimal

$(200.50)_{10} = ?$

☒ $(C8.8)_{16}$

☐ $(C4.4)_{16}$

☐ $(C6.6)_{16}$

☐ $(CA.A)_{16}$

↺ Unanswer

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21 ★ QUESTION: 26 GROUP: Programming Basics SECTION: C Programming Mark(s): 1

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What does the program given below do?

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```
#include <stdio.h>
int main()
{
    unsigned int num ;
    int c = 0;
    scanf("%u", &num);
    for (; num; num >= 1)
    {
        if (num & 1)
            c++;
    }
    printf("%d", c);
    return 0;
}
```

☒ It counts the number of bits which are on in the number num.

☐ It counts the number of bits which are off in the number num.

☐ It sets all bits in the number num to 0.

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22 QUESTION: 27 GROUP: Programming Basics SECTION: C Programming Mark(s): 1

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What will be the output of the program given below?

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```
#include <stdio.h>
int main()
{
    enum color
    {
        red,
        blue = -4,
        black = 3,
        white,
        green = 3,
        sky,
    };
    static int i, j;
    int a = white, b = sky, c = red;
    enum color col;
    col = black + blue / 2;
    for(; i <= col; i++)
    {
        printf("%d ", j = (col >= sky) ? (red >= b) ? (a >= b) ? 'a' : (c >= b < a) ? '4' : '3' : '1' : '2' );
        printf(" %d ", frut(col, j));
    }
}
```

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```
22 {
23     enum color
24     {
25         red,
26         blue = -4,
27         black = 3,
28         white,
29         green = 3,
30         sky,
31     };
32     static int i, j;
33     int a = white, b = sky, c = red;
34     enum color col;
35     col = black + blue / 2;
36     for( i <= col; i++)
37     {
38         printf("%d ", j = (col >= sky) ? (red >= b) ? (a >= b) ? 'a' : (c >= b < a) ? '4' : '3' : '1' : '2' );
39         printf(" %d ", frut(col, j));
40     }
41     return 0;
42 }
43
44 int frut(int s, int t)
45 {
46     static int val;
```

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```
22 {
23     printf("%d ", j = (col >= sky) ? (red >= b) ? (a >= b) ? 'a' : (c >= b < a) ? '4' : '3' : '1' : '2' );
24     printf(" %d ", frut(col, j));
25 }
26 return 0;
27 }
28
29 int frut(int s, int t)
30 {
31     static int val;
32     val += s++;
33     t = val++;
34     return t;
35 }
36
```

☐ 51 3 51 5☐ 48 2 49 4☐ 50 1 50 3☐ 52 1 53 4

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23 QUESTION: 28 GROUP: Programming Basics SECTION: Data Structures Basics Mark(s): 1

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25 ★

Consider two printers i.e. one is a monochrome printer and another one is a color printer. There are a series of pdf to print using these two printers. X indicates that the paper has to be printed by a monochrome printer and Y indicates that the paper has to be printed by color printer. Sometimes there is a chance that paper gets stuck in both printers it is denoted as Z and W for monochrome and color printers respectively. Whenever the paper gets stuck remove it from the printer. So given table denotes the series of printing operations and the number of characters to be printed.

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Consider the following table.

28 ☆

Sl.No.	Print Operation	No. of characters to print
1	X	30
2	X	40
3	Y	22
4	Y	34
5	Z	
6	W	
7	X	60
8	Y	67

29

Use two queues that are empty and of size 5. The printer is said to be efficient when the number of characters printed is more in the corresponding queue. Identify which printer is efficient.

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☐ Color Printer

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☐ Both Monochrome and Color Printer

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Consider the following table.

24

Sl.No.	Print Operation	No. of characters to print
1	X	30
2	X	40
3	Y	22
4	Y	34
5	Z	
6	W	
7	X	60
8	Y	67

25 ★

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28 ☆

Use two queues that are empty and of size 5. The printer is said to be efficient when the number of characters printed is more in the corresponding queue. Identify which printer is efficient.

29

☐ Color Printer

30

☐ Both Monochrome and Color Printer

31

☐ None of the options given

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☐ Monochrome Printer

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QUESTION: 21 GROUP: Programming Basics SECTION: Operating System Fundamentals Mark(s): 1

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Which one of the given option must be declared in banker's algorithm when a new process enters the system?

☐ The process must declare the maximum number of instances of each resource type that it may need.

☐ The process must declare the maximum number of resources that it holds.

☐ The process must declare the maximum number of resources that it may need.

☐ The process must declare the maximum number of instances of each resource type that it holds.

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QUESTION: 42 GROUP: Computer Science SECTION: C programming Mark(s): 1

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Which one of the statement given in options is true about the usage of structure?

☒ None of the given options

☐ Storage class can be assigned to an individual member

☐ The scope of a member name is confined to the particular structure, within which it is defined

☐ Individual members can be initialized within a structure type declaration

Unanswer

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41 QUESTION: 43 GROUP: Computer Science SECTION: C programming Mark(s): 1

42 What will be the output of the program given below?

43 ☆
44 #include <stdio.h>

45 int main()

46 {

47 char a[] = "abode";

48 char *p = a;

49 p++;

50 p++;

51 printf("%s", p);

52 return 0;

53 }

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☐ de☒ cde☐ bode☐ ade[Unanswer](#)[Previous Question](#)[Next Question](#)[End Test](#)

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41 QUESTION: 44 GROUP: Computer Science SECTION: C programming Mark(s): 1

42 What will be the output of the program given below?

43 #include <stdio.h>

44 int main()

45 {

46 int m = 3, n = 11, p;

47 p = (m++) ? m == 1 && n : 0;

48 printf("%d ", p);

49 return 0;

50 }

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☐ 2☐ 3☒ 0☐ 1[Unanswer](#)[Previous Question](#)[Next Question](#)[End Test](#)

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TIME LEFT: 00:06:58

QUESTION: 45 GROUP: Computer Science SECTION: Computer Architecture 1 Mark(s): 1

Which of the below given is/are involved in micro-operations?

I) Register Transfer
II) Arithmetic operation
III) Logic operation

☐ Only (I)

☐ Only (II)

☐ All (I), (II) and (III)

☒ Only (I) and (II)

[Unanswer](#)

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TIME LEFT: 00:06:57

QUESTION: 46 GROUP: Computer Science SECTION: OOPS Mark(s): 1

Consider the image given below:

```
graph BT; Dell --> Laptop; Laptop --> Object
```

Which of the type of relationships in Inheritance is representing the above diagram?

☐ USES - A

☐ HAS- A

☒ IS- A

☐ None of the given options

[Unanswer](#)

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TIME LEFT: 00:06:56

QUESTION: 47 GROUP: Computer Science SECTION: OOPS Mark(s): 1

Which of the below given statement(s) is/are rule(s) for Overriding a method?

I) If a method cannot be inherited, then it cannot be overridden.

II) A method declared as static cannot be overridden and it cannot be re-declared too.

III) Instance methods cannot be overridden.

☐ Only (III)

☒ Only (I)

☐ Only (I) and (III)

☐ Only (II) and (III)

Unanswer

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QUESTION: 49 GROUP: Computer Science SECTION: Computer Architecture 1 Mark(s): 1

A non – pipelined architecture runs at 5 GHz and it takes 10 clock cycles to finish the instruction. The machine runs at 1000 MHz and it has the stall caused by the memory and the branch. The stall caused by the memory has 40% of the total instructions and a stall of 4 cycles happens in 40% of the memory instructions. There is another stall caused by the branch which has 40% of the total instructions and a stall of 80 cycles happens in 3% of the memory instructions. What is the value of the CPI pipelined with the ideal CPI as 1?

☐ 9.6

☐ 2.6

☐ 11.6

☐ 6.4

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45 ★

QUESTION: 50 GROUP: Computer Science SECTION: Data Structures Mark(s): 1

46

Consider a complete binary tree with n nodes. What is the result of below given method test()?

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```
void test (struct Node *root)
{
    char *arr = new char[MAX]
    int index = 0
    storeAlternate (root, arr, &index, 0)
    reverse (arr, index)
    index = 0
    modifyTree(root, arr, &index, 0)
}
```

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```
void reverse(char arr[], int n)
{
    int l = 0, r = n - 1
    while (l < r)
    {
        int temp = arr[l]
        arr[l] = arr[r]
        arr[r] = temp
        l++
        r--
    }
}
```

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TIME LEFT: 00:06:52

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QUESTION: 51 GROUP: Computer Science SECTION: C programming Mark(s): 1

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To how many pointer variables we can assign address of an integer variable?

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☒ Any number of pointer variables

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☐ 1

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☐ 3

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☐ 2

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TIME LEFT: 00:06:51

QUESTION: 52 GROUP: Computer Science SECTION: Data Structures Mark(s): 1

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Consider the following sentences

Sentence 1: "Shell sort is mainly a variation of Insertion Sort".

Sentence 2: "In insertion sort, we move elements only one position ahead."

Sentence 3: "When an element has to be moved far ahead, many movements are involved."

Follow the below algorithm to get the modified text.

i) Remove all the words that exist identically in all three sentences.

ii) Remove the duplicate words in each sentence.

iii) Delete all words that begin with vowels.

iv) Insert all the words into the linear queue in the decreasing order of their length.If the length is same then discard such words.(Insertion Starts from Sentence 1 Sentence 3) (size of the linear queue is 6).

v) Dequeue all the words and identify the modified text from the following options.

☐ position mainly

☐ shell mainly variation sort we move position when Has to be moved far many movements

☐ movements many far moved be to Has when position move we sort variation mainly shell

☐ mainly position

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QUESTION: 53 GROUP: Computer Science SECTION: Operating System Mark(s): 1

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Consider a disk system has 14 recording surface with 1024 tracks per surface. The diameter of the inner cylinder is 8 inches and that of the outer cylinder is 12 inches, then what will be its track density?

☐ 1024 tracks/inch

☐ 2048 tracks/inch

☐ 256 tracks/inch

☐ 512 tracks/inch

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QUESTION: 54 GROUP: Computer Science SECTION: C programming Mark(s): 1

What will be the output of the program given below?

```
#include <stdio.h>

int result();

int fun(int, int*);

int main()
{
    int x = 15;

    printf("%d\n", fun(result(), &x));

    return 0;
}

int fun(int n, int *f_p)
{
    int t, f;

    if (n <= 1)
    {
        *f_p = 1;

        return 1;
    }

    t = fun(n - 1, f_p);

    f = t + *f_p;

    *f_p = t;

    return f;
}
```

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QUESTION: 55 GROUP: Computer Science SECTION: OOPS Mark(s): 1

Which of the below given statement(s) is/are the rule(s) for Method Overriding?

- I) The method must have the same parameter as in the parent class.
- II) It must be HAS-A relationship.
- III) The method must have the same name as in the parent class.

☒ Only (II)☐ Only (I) and (II)☐ Only (I) and (III)☐ Only (I)[Unanswer](#)

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QUESTION: 56 GROUP: Computer Science SECTION: Computer Architecture 1 Mark(s): 1

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56 ☆ Flag this question

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A J-K flip flop has its J input connected to logic level 1 and its input to the Q, output pulse is fed to its clock input, the flip flop will now:

☐ Retain its previous state

☐ Go to state 1 and stay there

☐ Change its state at each clock pulse

☐ Go to state 0 and stay there

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QUESTION: 41 GROUP: Computer Science SECTION: Algorithms Mark(s): 1

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What will be the output of the program given below?

Algorithm:

1. Start

2. Initialize x, y = 20, z = 15

3. Set x = y ÷ 2

4. Check if 0 == x then

4a. x = 10 ÷ 3

5. Set i = 1

6. Repeat 6a to 10 for i <= x step by 1

6a. Print i + 2

7. Set j = 1

8. Repeat 9 to 10 for j <= 5 step by 1

9. Set z = x ÷ y;

10. Print "ABC"

End for

11. Set j = j + 1

12. Print j + 1

End if

13. Stop

☐ ABC 10

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TIME LEFT: 00:06:45

QUESTION: 58 GROUP: Computer Science SECTION: Data Structures Mark(s): 1

Consider:

```
bool Test(Node *root, Node *left = NULL, Node *right = NULL)
{
    if (root == NULL)
        return true;

    if (left != NULL and root->data < left->data)
        return false;

    if (right != NULL and root->data > right->data)
        return false;

    return Test(root->left, left, root) and Test(root->right, root, right);
}
```

What task does the above function perform?

☐ Checks if the tree is an AVL Tree.☐ Checks if the tree is a Complete Binary Tree.

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QUESTION: 59 GROUP: Computer Science SECTION: C programming Mark(s): 1

What will be the output of the program given below?

```
#include <stdio.h>
int main()
{
    int a = -11, b = 3, c = 5;
    register int m, s, i;
    m = a++ % b++ / 2 + c--;
    s = ++m + s++;
    for (i = 0; i < 3; trace(m++ + s-- / 2))
    {
        printf("%d ", m >> 2 + s);
        break;
    }
    return 0;
}

int trace(int l)
{
    printf(" %d", l--);
    return(++l / 2);
}
```

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QUESTION: 60 GROUP: Computer Science SECTION: Operating System Mark(s): 1

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48 ★ Which of the statement given in options is not true regarding safe/unsafe state of a system?

49 ☐ An unsafe state may lead to a deadlock.

50 ☐ A deadlocked state is an unsafe state.

51 ☐ A safe state is not a deadlocked state.

52 ☒ All unsafe states are deadlocks.

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54 [Unanswer](#)

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TIME LEFT: 00:06:46

QUESTION: 57 GROUP: Computer Science SECTION: C programming Mark(s): 1

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48 ★ What will be the output of the program given below?

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```
#include <stdio.h>
union abed
{
    int a;
    char b;
};
struct abce
{
    int a;
    char b;
};
int main()
{
    union abed u = {10};
    struct abce *pa = (struct abce *)&u;
    union abed *pu = (union abed *)&u;
    printf("%d\n", pa->a);
    printf("%d %d\n", pu->a, pu->b);
    return 0;
}
```

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60 ☐ 10

60 ☐ 10 80

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